FDA REGULATORY ESSENTIALS

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Drugs · Medical Devices · Biologics · Post-Market · Regulatory Strategy

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Build Projects – FDA Regulatory Essentials Fellowship

April 2026

Table of Contents

Table of Contents 2
Chapter 1: Investigational New Drug Application (IND) 5
1.1 The Three Types of IND Application 5
1.1.1 Commercial IND 5
1.1.2 Investigator IND 5
1.1.3 Emergency IND (Compassionate Use / Expanded Access) 6
Chapter 2: New Drug Application (NDA) 6
2.1 Full NDA (Standard and Priority Review) 6
2.2 505(b)(2) Hybrid NDA 6
2.3 Supplemental NDA (sNDA) 7
Chapter 3: Abbreviated New Drug Application (ANDA) 7
3.1 ANDA Patent Certification Paragraphs 7
3.2 What an ANDA Must Contain 8
Chapter 4: Clinical Trial Phases — The IND Pathway Timeline 8
Chapter 5: eCTD and ESG — Format vs. Portal 10
5.1 eCTD: The Format (The Filing Cabinet) 10
5.2 ESG: The Portal (The Delivery Truck) 10
5.3 The Five eCTD Modules in Detail 11
Chapter 6: 21 CFR Part 11 — Electronic Records and Signatures 11
6.1 What Part 11 Covers 11
6.2 The Four Core Requirements 12
6.2.1 Unique User IDs 12
6.2.2 Access Controls 12
6.2.3 Audit Trails 12
6.2.4 Record Retention 12
Chapter 7: Orange Book and DRUGS@FDA 12
7.1 The Orange Book (orangebook.fda.gov) 12
7.1.1 Therapeutic Equivalence (TE) Codes 13
7.1.2 How to Navigate the Orange Book 13
7.2 DRUGS@FDA (accessdata.fda.gov/scripts/cder/daf) 13
Chapter 8: QMSR and the 21 CFR Regulatory Framework 14
8.1 Key 21 CFR Parts Reference 14
8.2 QMSR (21 CFR Part 820) vs ISO 13485 14
Chapter 9: Medical Device Classification and Regulatory Pathways 15
9.1 The Three Risk Classes 15

9.2 Premarket Pathways 16
9.2.1 510(k) — Substantial Equivalence 16
9.2.2 De Novo — The Pioneer Pathway 16
9.2.3 PMA — Premarket Approval 16
9.3 Common Device Regulatory Pitfalls 17
9.3.1 The Split Predicate 17
9.3.2 Intended Use Drift 17
9.3.3 Wrong Product Code Selection 17
9.3.4 Missing or Weak Predicate Bridge 17
Chapter 10: Biologics, Biosimilars, and the BLA 18
10.1 Small Molecules vs Biologics vs Biosimilars 18
10.2 BLA 351(a) — The Innovator Pathway 18
10.3 BLA 351(k) — The Biosimilar Pathway (Stepwise Approach) 18
10.4 Biologic Exclusivity and the Patent Dance 19
Chapter 11: Quality Management Systems (QMS) 20
11.1 The Six Core QMS Pillars 20
11.1.1 Document Control 20
11.1.2 Change Control 20
11.1.3 CAPA — Corrective and Preventive Action 20
11.1.4 Training Management 20
11.1.5 Deviation Management 21
11.1.6 Audit and Inspection Readiness 21
Chapter 12: Post-Market Surveillance — Report, Record, Recall 21
12.1 Report — Adverse Event Capture 21
12.2 Record — Triage and Complaint File Management 21
12.3 Recall — Market Withdrawal 22
Chapter 13: FAERS and MAUDE — The Post-Market Databases 22
13.1 FAERS — FDA Adverse Event Reporting System 22
13.2 MAUDE — Manufacturer and User Facility Device Experience 23
Chapter 14: Language Hygiene — Writing Like a Regulatory Professional 24
14.1 Drug Adverse Event (MedWatch / FAERS) 24
14.2 Device Malfunction (MDR / MAUDE) 24
14.3 510(k) Predicate Comparison 25
14.4 Recall / Field Safety Notice 25
Chapter 15: Regulatory Strategy — Connecting Commercial Goals to Evidence 26
15.1 The Unified Pathway Matrix 26

Commercial Goal 26
Development Risk Bridge 26
Regulatory Requirement 26
15.2 Expedited FDA Programmes for Unmet Medical Needs 26
15.3 Centre-Specific Regulatory Playbooks 27
CDER (Center for Drug Evaluation and Research) — Drugs 27
CBER (Center for Biologics Evaluation and Research) — Biologics 27
CDRH (Center for Devices and Radiological Health) — Medical Devices 28
Key Regulatory References and Resources 28

Chapter 1: Investigational New Drug Application (IND)

Before most clinical investigations of an unapproved drug begin in the United States, the sponsor must submit an IND unless an exemption applies. The investigation generally may proceed 30 days after FDA receives the IND, unless FDA places it on clinical hold, or earlier if FDA notifies the sponsor it may proceed. IRB approval and other applicable requirements also apply.

Understanding the IND is foundational to drug development because it defines who is accountable, what data must be on file, and what oversight structures must be in place before a single human subject is enrolled.

1.1 The Three Types of IND Application

1.1.1 Commercial IND

The Commercial IND is submitted by pharmaceutical or biotechnology companies that intend to develop a drug for commercial sale. This pathway leads ultimately to a New Drug Application (NDA). It carries the most rigorous requirements because the sponsor anticipates widespread public exposure to the product.

Key requirements include:

- A full pre-clinical data package covering pharmacology and toxicology
- A Chemistry, Manufacturing and Controls (CMC) section detailing how the drug is made and its quality specifications
- A detailed investigational plan including clinical protocols for each proposed study
- Investigator qualifications and a copy of the Institutional Review Board (IRB) approval letter
- Pharmacology and toxicology summary documents

> Real-world example: A biotech company developing a novel JAK inhibitor for rheumatoid arthritis submits a Commercial IND. They include 18 months of rat and monkey toxicology data, GMP-manufactured drug substance specifications, and a Phase 1 protocol targeting 24 healthy volunteers in a dose-escalation design.

1.1.2 Investigator IND

The Investigator IND is used by academic researchers or treating physicians who initiate and take full responsibility for a clinical trial. In this case, the sponsor and the investigator are the same person or institution. This pathway is commonly used in oncology and rare disease research, often to repurpose already-approved drugs for new indications.

Compared to the Commercial IND, the data package is simpler because the investigator is not seeking commercial approval; they are generating scientific evidence. However, the investigator still assumes full regulatory responsibility.

> Real-world example: A university oncologist wants to test low-dose naltrexone (an FDA-approved drug) as an adjunct therapy in patients with multiple sclerosis. Since naltrexone is already approved and the investigator is initiating the trial themselves, they file an Investigator IND. They are both the sponsor and the investigator on the FDA's records.

1.1.3 Emergency IND (Compassionate Use / Expanded Access)

The Emergency IND, also called a Compassionate Use IND or Expanded Access IND, is used when a physician needs to treat a single patient or a small group of patients with a life-threatening condition using an unapproved drug, and there is no satisfactory alternative treatment available.

This is a critical distinction: the Emergency IND is not a Fast Track or Breakthrough Therapy designation. Those are expedited review programmes for drugs already in clinical development. The Emergency IND is a separate, urgent access mechanism.

- The FDA can authorize the Emergency IND verbally within hours
- A written follow-up must be submitted within 15 working days
- It is also accessed through the Expanded Access Program (EAP) formal pathway

> Real-world example: A critically ill COVID-19 patient requires remdesivir before it had received full approval. The treating physician contacts the FDA by phone, receives verbal authorization for an Emergency IND, administers the drug, and files the written follow-up within the required 15 days.

Chapter 2: New Drug Application (NDA)

The New Drug Application (NDA) is the master regulatory submission that a sponsor files with the FDA to obtain permission to market a drug product commercially in the United States. Governed by 21 CFR Part 314, the NDA must compile all evidence demonstrating that the drug is safe, effective, and consistently manufactured. There are three distinct NDA pathways, each suited to a different commercial context.

2.1 Full NDA (Standard and Priority Review)

A 505(b)(1) NDA contains full reports of investigations of safety and effectiveness, generally conducted by or for the applicant or supported by a right of reference. This pathway is not limited to brand-new molecular entities. Standard and priority review are FDA review designations, not separate statutory NDA types.

- Module 1: Administrative — labeling, forms, cover letters (region-specific)
- Module 2: Summaries — the reviewers' entry point; a strong Module 2 is critical to first impressions
- Module 3: CMC — Chemistry, Manufacturing and Controls covering the drug substance and product
- Module 4: Non-clinical reports — animal pharmacology, toxicology, and PK/PD studies
- Module 5: Clinical study reports — Phases 1 through 3 human trial data

FDA's review timeline is governed by the Prescription Drug User Fee Act (PDUFA). Standard Review has a 10-month PDUFA goal. Priority Review for drugs treating serious conditions with unmet medical need has a 6-month goal.

2.2 505(b)(2) Hybrid NDA

The 505(b)(2) is a hybrid pathway that allows the sponsor to rely, at least in part, on published literature or existing FDA findings of safety and efficacy data the sponsor does not own. This

avoids repeating full Phase 1-3 clinical programmes where existing evidence already supports the new product's safety profile.

Common 505(b)(2) scenarios include:

- New formulations of an approved active ingredient (e.g., a new extended-release version)
- New dosage forms (e.g., switching from tablet to patch)
- New routes of administration of an already-approved drug
- New fixed-dose combinations of two approved drugs
- New indications supported by published literature

> Real-world example: A company develops a novel nasal spray formulation of sumatriptan (already approved as a tablet for migraines). They file a 505(b)(2) NDA, referencing the existing tablet NDA for pharmacological evidence and adding their own bioavailability and local tolerance studies for the new route of administration.

2.3 Supplemental NDA (sNDA)

After an NDA is approved, postapproval changes follow the applicable reporting categories under 21 CFR 314.70. Depending on the change, the applicant may submit a prior-approval supplement, a changes-being-effected supplement, or information in an annual report. Not every change requires a supplemental NDA.

- Prior-Approval Supplement (PAS): Major changes requiring FDA approval before implementation, such as a new indication, a new manufacturing site, or a significant labeling change
- CBE-30 (Changes Being Effected in 30 Days): Moderately significant changes; the sponsor notifies FDA 30 days before implementation
- CBE-0 (Changes Being Effected Immediately): Minor changes that can be made immediately; notice is submitted concurrently
- Annual Report: Very minor changes documented in an annual summary to FDA

The sNDA is not a one-time event. It is used continuously throughout the product lifecycle for manufacturing site changes, label updates, new paediatric indications, and safety revisions triggered by post-market data.

Chapter 3: Abbreviated New Drug Application (ANDA)

The ANDA pathway was created by the Hatch-Waxman Act of 1984 to enable the approval of generic drugs in the United States. Rather than repeating full clinical trials, generic drug applicants must demonstrate that their product is bioequivalent to the Reference Listed Drug (RLD), the FDA-approved innovator product against which the generic is compared. Today, approximately 90% of all dispensed prescriptions in the United States are filled with generics approved via the ANDA pathway.

3.1 ANDA Patent Certification Paragraphs

The most strategically significant aspect of an ANDA is the patent certification declaration, required under the Hatch-Waxman Act. The applicant must certify under one of four paragraphs:

- Paragraph I: No patent has been listed for the reference listed drug in the Orange Book. The applicant can proceed without litigation risk.
- Paragraph II: The listed patent(s) have already expired. The applicant can proceed immediately.
- Paragraph III: The listed patent will expire on a specific date. The ANDA is approved but becomes effective only after that expiry date.
- Paragraph IV: The applicant asserts that the listed patent is either invalid or will not be infringed by the generic. This is the most aggressive certification it can trigger a 30-month stay of approval while the patent holder litigates. However, the first ANDA filer to certify Paragraph IV on a particular brand drug earns 180 days of generic market exclusivity.

Additionally, a Suitability Petition ANDA is available when the proposed generic differs from the RLD in strength, dosage form, or route of administration. The applicant must first petition FDA to confirm that the ANDA pathway is appropriate.

3.2 What an ANDA Must Contain

Section	Requirement
Bioequivalence Studies	PK study demonstrating AUC and Cmax within 80–125% of RLD (using a 90% Confidence Interval). Both parameters must meet this range.
Labeling	Must match the RLD label exactly, except for permitted carve-outs (e.g., patented indications may be removed).
CMC — Module 3	Drug substance and product specifications, manufacturing process, container-closure system, and stability data.
Patent Certification	Para I, II, III, or IV declaration under the Hatch-Waxman Act.
Inactive Ingredient Compliance	All excipients must be within limits documented in the FDA Inactive Ingredient Database (IIG).
Site Registration	Manufacturing facility DUNS and FEI numbers required. Facility must be registered and listed.

> Real-world example: A generic manufacturer wants to produce atorvastatin calcium 10mg tablets (generic for Lipitor). They search the Orange Book, identify Lipitor as the RLD, confirm the AB TE code, find that the main patent expired in 2011, certify Paragraph II, conduct a 24-subject crossover bioequivalence study showing 96% AUC ratio and 98% Cmax ratio vs Lipitor, and submit their ANDA. This is now the most-prescribed drug in the US in generic form.

Chapter 4: Clinical Trial Phases — The IND Pathway Timeline

Once an IND is active, the drug enters a structured sequence of clinical development phases. Each phase has a distinct purpose, patient population, and success threshold. The failure rate is significant at each stage: one widely cited industry cohort (BIO/QLS/Informa, 2011–2020)

estimated a 7.9% likelihood of approval from Phase I, with marked variation across therapeutic areas and modalities.

Phase	Population	Duration	Purpose	Outcome
Pre-IND research	N/A: nonclinical	Variable	Nonclinical safety and development	N/A
Phase 1	Usually 20–80	1–2 years	Safety, dose, and pharmacokinetics	52.0% advance*
Phase 2	Usually 100–300	~2 years	Preliminary efficacy, dose and safety	28.9% advance*
Phase 3	Often 1,000+	2–4 years	Confirmatory efficacy and safety	57.8% advance*
NDA submission	N/A	Variable	Submit drug application and evidence	Not a trial stage
ANDA submission	N/A	Variable	Submit generic application and evidence	Not a trial stage
FDA NDA review	N/A	10 months standard; 6 priority**	Review application and labeling	Varies
FDA ANDA review	N/A	Generally 10 months**	Review generic application	Varies
Post-market monitoring	Marketed-product users	Ongoing	Monitor safety and required follow-up	N/A

* Industry phase-transition estimates from BIO/QLS Advisors/Informa, Clinical Development Success Rates 2011–2020 (published 2021): Phase I 52.0%, Phase II 28.9%, Phase III 57.8%. These are cohort observations, not FDA-mandated success rates and not forecasts for a specific program. ** FDA review goals are action goals, not promises of approval.

Postmarketing pharmacovigilance and applicable reporting duties continue after approval. However, a separate Phase 4 clinical study is not mandatory for every drug. FDA may impose postmarketing requirements (PMRs), and sponsors may undertake postmarketing commitments (PMCs); these categories have different legal implications.

> Note on ANDA: The ANDA pathway entirely bypasses Phases 1–3. Instead of conducting clinical trials, the generic applicant performs a pharmacokinetic bioequivalence study (typically 24–36 subjects) comparing the generic to the RLD under fasting and fed conditions. If the PK parameters fall within the 80–125% range with 90% confidence, the product is considered bioequivalent.

Chapter 5: eCTD and ESG — Format vs. Portal

One of the most common sources of confusion among regulatory affairs newcomers is the relationship between the eCTD and the ESG. These are two distinct things that work together, but confusing them leads to submission errors.

5.1 eCTD: The Format (The Filing Cabinet)

The Electronic Common Technical Document (eCTD) is the globally harmonized, standardized format for organizing and submitting drug regulatory dossiers to health authorities, including the FDA, EMA, Health Canada, and PMDA (Japan). It is not a piece of software — it is a structured folder architecture with an XML backbone that links, indexes, and enables navigation of thousands of documents.

- Globally harmonized under the ICH M4 guideline
- eCTD is the structured electronic submission standard for submissions to which FDA's eCTD requirements apply. Applicability and any exceptions depend on submission type and the effective requirements. Confirm current technical specifications before preparing a dossier.
- Uses an XML backbone to link and index all component documents
- Enables FDA reviewers to electronically navigate the dossier
- Supports lifecycle management: each new submission builds on the previous one
- Module 1 is region-specific; Modules 2 through 5 are globally harmonized

> Critical point: The eCTD is the archive the folder structure containing all your documents. It does not send itself. It must be transmitted through a separate system: the ESG.

5.2 ESG: The Portal (The Delivery Truck)

The FDA Electronic Submissions Gateway Next Generation (ESG NextGen) securely receives and routes regulatory submissions. The Unified Submission Portal (USP) replaced the legacy WebTrader interface on April 14, 2025. Industry users should follow current FDA submission-type guidance and the applicable USP, API, or AS2 workflow.

- FDA's modernized electronic submission gateway; follow product- and center-specific submission guidance.
- ESG NextGen provides updated submission infrastructure; do not assume historical WebTrader instructions remain current.
- Access, identity verification, onboarding, and security requirements depend on the active ESG NextGen workflow.
- Applicable submissions undergo technical and regulatory validation checks; successful transmission alone does not establish acceptance for review.
- Submission routes and technical standards differ by product type and FDA center; not all device submissions use eCTD.
- Electronic-record controls must be assessed under applicable FDA requirements, including 21 CFR Part 11 where relevant; platform security does not by itself establish a sponsor's compliance.

> The workflow: (1) Build your eCTD dossier using a validated authoring tool. (2) Run it through an eCTD validation tool to check compliance. (3) Log into the ESG portal. (4) Upload and transmit your validated eCTD package.

5.3 The Five eCTD Modules in Detail

Module	Title	Scope	Key Contents
M1	Administrative & Regional	Region-specific (differs by country)	Cover letters, submission forms, proposed labeling (package insert), patent certifications (Hatch-Waxman), regional administrative requirements
M2	Summaries & Overviews	Global — identical worldwide	Quality Overall Summary (QOS), non-clinical overview and tabular summaries, clinical overview and written summaries. Reviewers read this first a strong M2 signals a well-organized dossier.
M3	Quality (CMC)	Global — identical worldwide	Drug substance characterization and synthesis, drug product composition and manufacturing, container-closure system, analytical methods, validation data, and stability studies
M4	Non-Clinical Study Reports	Global — identical worldwide	Pharmacodynamics (mechanism of action), pharmacokinetics ADME in animal models, toxicology (single dose, repeat dose, genotoxicity, carcinogenicity), reproductive and developmental toxicology
M5	Clinical Study Reports	Global — identical worldwide	Phase 1–3 clinical study reports, individual patient data and listings, efficacy and safety tabular summaries, post-market data where applicable

Chapter 6: 21 CFR Part 11 — Electronic Records and Signatures

21 CFR Part 11 applies to electronic records created, modified, maintained, archived, retrieved or transmitted under applicable FDA record requirements, and to electronic signatures intended to be equivalent to handwritten signatures, within its defined scope. Applicability is not automatic for every digital document or electronic system.

The regulation exists because the FDA recognized, as the industry moved from paper to digital systems, that electronic records required the same level of trustworthiness, reliability, and integrity as paper records. Part 11 establishes the controls that provide that assurance.

6.1 What Part 11 Covers

- Clinical trial data and Electronic Data Capture (EDC) systems
- Electronic lab notebooks (ELN) used in drug substance and product development
- Manufacturing batch records and production control systems

- Quality management systems (QMS) including CAPA, deviation, and change control records
- eCTD submission documents transmitted via ESG

6.2 The Four Core Requirements

6.2.1 Unique User IDs

For Part 11-covered closed systems, access must be limited to authorized individuals and electronic signatures must be attributable to the individual signer. User access and signature controls should be assessed against the specific applicable requirements; shared accounts can undermine attribution and data integrity.

> Example violation: Two laboratory technicians share a single login ("labtechuser") to log analytical results. This makes it impossible to determine which technician entered or modified specific data, constituting a direct Part 11 violation. The FDA routinely finds and cites this during inspections.

6.2.2 Access Controls

Regulated electronic systems must implement role-based access controls. This means that each user's access is limited to exactly what their role requires nothing more. A quality assurance reviewer should be able to view and approve records but not alter original data. A laboratory technician should be able to enter results but not approve or release product. Segregation of duties must be enforced systematically, not just by policy.

6.2.3 Audit Trails

For Part 11-covered closed systems, secure, computer-generated, time-stamped audit trails are required for covered actions involving creation, modification, or deletion of electronic records. Audit trails should be retained and available for FDA review as required; claims that every interaction must be recorded are overly broad.

6.2.4 Record Retention

Electronic records must remain accurate, retrievable, and readable for the duration required by the applicable regulation often long after the drug has been approved or discontinued. A record that cannot be retrieved in readable form at the time of an FDA inspection effectively does not exist from a regulatory standpoint.

Chapter 7: Orange Book and DRUGS@FDA

Two of the most powerful and frequently underutilised FDA databases are the Orange Book and DRUGS@FDA. They serve fundamentally different purposes and are used at different stages of the regulatory intelligence process. Experienced regulatory professionals use both routinely.

7.1 The Orange Book (orangebook.fda.gov)

The Orange Book lists FDA-approved drug products within its defined scope and therapeutic-equivalence evaluations for eligible multisource prescription products. It is an important resource

for regulatory intelligence, including generic drug development, but it is not a complete database of every FDA-regulated medicine.

7.1.1 Therapeutic Equivalence (TE) Codes

Every product in the Orange Book carries a TE code, which signals whether the product is considered therapeutically equivalent to the RLD. The codes matter enormously for generic drug strategy:

Code	Meaning	Practical Implication
AB	Proven bioequivalent; meets bioequivalence standards	Gold standard. Pharmacists can freely substitute for the brand product without prescriber authorization.
AA	No known or suspected bioequivalence problems	Approved without a BE study requirement. Generally safe to substitute.
BX	Insufficient data to determine equivalence	Use with caution. More data may be needed before substitution is appropriate.
BC	Extended-release dosage forms without adequate demonstration of equivalence	B-rated: not considered therapeutically equivalent on the available evidence; adequate bioequivalence data can support an AB rating.
BE	Delayed-release oral dosage forms without adequate evidence of equivalence	B-rated: not considered therapeutically equivalent unless appropriate evidence supports a different classification.

7.1.2 How to Navigate the Orange Book

1. Go to orangebook.fda.gov and search by active ingredient (e.g., ‘atorvastatin’)
2. Review the results to identify the Reference Listed Drug (marked ‘RLD’) and the Reference Standard (RS)
3. Read the TE Code to determine whether existing generics have been deemed therapeutically equivalent
4. Click the NDA number to view all associated patent listings and exclusivity expiry dates
5. Based on patent status, determine whether a Paragraph I, II, III, or IV certification is appropriate
6. Use exclusivity expiry data to plan ANDA submission timing and commercial launch strategy

7.2 DRUGS@FDA (accessdata.fda.gov/scripts/cder/daf)

While the Orange Book tells you what has been approved and when patents expire, DRUGS@FDA tells you how and why each product was approved — in the FDA’s own words. It provides public access to every FDA drug approval action, including the FDA’s own review documents.

What you can access:

- Official approval letters with exact approval dates and any post-market commitments (PMCs/PMRs)
- Current and historical prescribing information, package inserts, and medication guides

- Full FDA review documents: clinical, statistical, clinical pharmacology, and biopharmaceutics reviews
- Risk Evaluation and Mitigation Strategy (REMS) documents for high-risk drugs
- Complete bioequivalence review documents for ANDA products
- Complete Response Letters (CRLs) FDA's explanations for non-approval decisions

> Strategic tip: The most powerful use of DRUGS@FDA is to download and study your competitors' FDA review documents. The FDA's medical review of a competing NDA tells you exactly what evidence standard FDA applied, what issues they raised, what they required for approval, and what post-market commitments they imposed. This intelligence is invaluable when designing your own dossier. All of it is publicly available.

Chapter 8: QMSR and the 21 CFR Regulatory Framework

The regulatory framework for pharmaceutical and medical device quality systems is built on a hierarchy of regulations found in Title 21 of the US Code of Federal Regulations (21 CFR). Understanding which parts apply to which product types is essential foundational knowledge for any regulatory affairs professional.

8.1 Key 21 CFR Parts Reference

21 CFR Part	Subject	Applies to
Part 314	NDA regulations	New drug applications and related procedures
Part 320	Bioavailability and Bioequivalence	BE requirements for drug products; underpins ANDA pathway
Parts 600s	Biologics regulations	Biologics License Applications (BLA) and biologic product standards
Part 820 (QMSR)	Quality Management System Regulation	Medical device manufacturers selling in the US
Part 11	Electronic records and signatures	All regulated environments using electronic systems
Part 807	Premarket notification	510(k) submissions for medical devices
Part 814	Premarket Approval (PMA)	Class III medical device approval applications

8.2 QMSR (21 CFR Part 820) vs ISO 13485

FDA's Quality Management System Regulation (QMSR), codified in 21 CFR Part 820, became effective on February 2, 2026. It incorporates ISO 13485:2016 by reference while retaining FDA-specific requirements. ISO 13485 certification and an FDA inspection are different processes;

certification alone does not establish compliance with all applicable US requirements. FDA inspections assess applicable regulatory compliance and do not award ISO certification.

Aspect	QMSR (21 CFR 820)	ISO 13485
Authority	US FDA mandatory for US market	International standard required by EU MDR, Canada MDSAP, Japan
Certification body	FDA inspects for compliance; it does not issue ISO 13485 certificates	Accredited third-party certification body (where certification is pursued)
Geographic scope	United States	EU, Canada, Japan, Australia, Brazil, and others
2026 effective date	QMSR effective February 2, 2026; ISO 13485:2016 incorporated by reference	ISO 13485:2016 remains a distinct standard; US-specific requirements still apply

Both QMSR and ISO 13485 require the same core quality system elements:

- Design controls (Design History File / DHF) documenting design inputs, outputs, reviews, verification, and validation
- CAPA (Corrective and Preventive Action) systematic root cause elimination
- Document and record control
- Supplier and purchasing controls
- Production and process controls
- Post-market surveillance

Chapter 9: Medical Device Classification and Regulatory Pathways

Medical devices in the United States are regulated by the Center for Devices and Radiological Health (CDRH) within the FDA. Unlike drugs, which are assessed primarily through clinical evidence, devices are classified into one of three risk classes that determine both the level of regulatory scrutiny and the premarket submission pathway. Approximately 43% of devices are Class I, 43% are Class II, and 10% are Class III.

9.1 The Three Risk Classes

Class	Risk Level	Regulatory Pathway	Examples
Class I	Low Risk (FDA illustrative estimate: 47%)	Exempt from premarket notification / General Controls only	Bandages, elastic bandages, tongue depressors, manual surgical instruments, non-sterile examination gloves
Class II	Low Risk (FDA illustrative	510(k) Premarket Notification (some	Infusion pumps, catheters, guidewires, ultrasound imaging systems, powered blood pressure cuffs, reusable syringes

	estimate: 47%)	exempt; De Novo for novel devices)	
Class III	Low Risk (FDA illustrative estimate: 47%)	PMA — Premarket Approval (most rigorous)	Prosthetic heart valves, implantable cardiac pacemakers, implantable defibrillators, cochlear implants, silicone breast implants

9.2 Premarket Pathways

9.2.1 510(k) — Substantial Equivalence

The 510(k) is the most common premarket submission pathway for Class II devices. The core concept is substantial equivalence (SE): the applicant must demonstrate that the new device has the same intended use as a legally marketed predicate device, and the same or substantially similar technological characteristics. If technological characteristics differ, the applicant must show that those differences do not raise new questions of safety or effectiveness.

Important terminology distinction: A 510(k) submission results in FDA 'clearance', not 'approval.' These are legally distinct terms. A PMA results in FDA 'approval.' FDA clears approximately 3,500 devices per year via the 510(k) pathway.

> Real-world example: A company develops a new powered infusion pump. They search the FDA 510(k) database and identify a cleared pump with the same clinical indication (controlled IV drug delivery to hospitalized patients) and similar microprocessor-controlled delivery mechanism. They submit a 510(k) with a predicate comparison table, bench testing data showing equivalent accuracy and alarm performance, and a software description. FDA reviews and issues a clearance letter within approximately 90 days.

9.2.2 De Novo — The Pioneer Pathway

The De Novo pathway is designed for genuinely novel devices that are low-to-moderate risk (Class I or Class II) but have no valid predicate in the cleared device database. When a device is so innovative that no legally marketed comparator exists, the De Novo pathway allows the applicant to establish a new product classification and create a product code that future 510(k)s can reference as a predicate.

> Real-world example: The first FDA-authorized software that uses artificial intelligence to detect diabetic retinopathy from fundus photographs was cleared via De Novo. The FDA cannot compare AI diagnostic software to a human ophthalmologist; there is no existing cleared predicate for this type of autonomous AI decision-making. The De Novo review established a new regulatory framework and product code for this category, enabling future similar software to reference it as a 510(k) predicate.

9.2.3 PMA — Premarket Approval

Premarket Approval (PMA) applies to devices for which that statutory pathway is required, generally Class III devices. An application must provide valid scientific evidence supporting reasonable assurance of safety and effectiveness. FDA may consider well-controlled investigations, other clinical evidence, and relevant supporting data as appropriate to the specific device. Randomized controlled trials can be important but are not universally mandatory.

PMA is an FDA approval pathway with a demanding evidence standard; other device marketing pathways also carry substantial controls and postmarket obligations. Avoid interpreting the prevalence of device classes as a reliable percentage of submissions requiring PMA.

- Prosthetic heart valves
- Implantable cardiac pacemakers and defibrillators
- Cochlear implants
- Total artificial hearts

A Modular PMA is available for complex Class III devices: the sponsor can submit individual modules (manufacturing, non-clinical studies, clinical data) sequentially as they are completed, rather than waiting until the entire dossier is assembled.

9.3 Common Device Regulatory Pitfalls

9.3.1 The Split Predicate

FDA describes an impermissible 'split predicate' approach as relying on one device to establish the same intended use and a different device with another intended use to establish technological characteristics. FDA generally must be able to complete the substantial-equivalence decision using one predicate, but additional predicate devices may support a submission in certain circumstances, including combinations of features sharing the same intended use. Multiple predicates are not categorically prohibited.

> ⚠ If no single predicate can support both the intended use and the technological characteristics of your device, the correct pathway is De Novo, not 510(k).

9.3.2 Intended Use Drift

A difference in indications, patient population, or setting may affect whether the new device has the same intended use as its proposed predicate. The substantial-equivalence analysis is fact-specific: not every change in indication automatically rules out 510(k), but a different intended use can. Compare the proposed labeling, risks, technological characteristics, and relevant FDA guidance before selecting a pathway.

> Example: A pulse oximetry device cleared for hospital use in adults cannot be directly 510(k)'d into pediatric home monitoring by citing the adult hospital device as a predicate. The intended population and use environment have changed significantly. A new substantial equivalence argument must be built, or De Novo/PMA may apply.

9.3.3 Wrong Product Code Selection

Every FDA device product code corresponds to a specific FDA review branch, regulation, and performance standards. Selecting a broad or incorrect product code routes the submission to the wrong review team and triggers inappropriate performance standards. The correct approach is to search the FDA Product Classification Database for the most specific product code that matches your device's intended use, and to confirm the selected code in a Pre-Submission (Q-Sub) meeting with FDA before filing.

9.3.4 Missing or Weak Predicate Bridge

When a device's technological characteristics differ from those of the predicate, the applicant must affirmatively demonstrate that those differences do not raise new questions of safety or effectiveness. This requires bench testing data, performance comparisons, and risk analysis. An

inadequate predicate bridge results in an Additional Information (AI) request from FDA, which extends the review timeline by at least 180 days.

Chapter 10: Biologics, Biosimilars, and the BLA

Biologics are a fundamentally different category of therapeutic product from small molecule drugs. This difference is not merely scientific, it has profound regulatory implications, driving an entirely separate approval pathway (the Biologics License Application, or BLA) and a different competitive landscape built around biosimilars rather than traditional generics.

10.1 Small Molecules vs Biologics vs Biosimilars

Characteristic	Small Molecule Drug	Biologic	Biosimilar
Size	~21 atoms; simple chemical structure	~25,000 atoms; large, complex 3D structure	Same large complex structure as reference
Origin	Chemically synthesized	Grown in living cell lines (human, animal, microbial)	Grown in living cell lines
Reproducibility	Identical copy possible: generic = exact replica	Impossible to perfectly replicate	Highly similar — not identical
Regulatory pathway	NDA (ANDA for generics)	BLA 351(a) — innovator pathway	BLA 351(k) — abbreviated biosimilar pathway
Approval equivalent	Generic = AB-rated via ANDA	No direct equivalent; biosimilar is not a generic	Interchangeable = highest designation
Reference database	Orange Book	Purple Book	Purple Book

10.2 BLA 351(a) — The Innovator Pathway

A Biologics License Application under section 351(a) of the Public Health Service Act is generally used for innovator biologics. FDA evaluates evidence that the product is safe, pure, and potent, including manufacturing and clinical evidence appropriate to the product. Development programs differ, and a fixed sequence of Phase 1, 2, and 3 studies should not be portrayed as universally mandatory. Review clocks and filing milestones should be confirmed against current FDA PDUFA guidance.

Critically, on approval, the 351(a) biologic becomes the ‘reference product’ against which all future biosimilar 351(k) applications must be compared.

10.3 BLA 351(k) — The Biosimilar Pathway (Stepwise Approach)

The biosimilar pathway, established under Section 351(k) of the Public Health Service Act and operationalised by the Biologics Price Competition and Innovation Act (BPCIA), allows a biosimilar applicant to rely on the safety and efficacy data generated by the reference product.

The biosimilar does not need to prove from scratch that it works it needs to prove that it is 'highly similar' to the reference product with no clinically meaningful differences.

The stepwise approach to demonstrating biosimilarity begins with the most sensitive data:

1. Analytical Studies: Structural and functional characterisation using advanced techniques including mass spectrometry, glycan mapping, and bioassays. This is the most important and informative step. Strong analytical similarity can reduce or waive subsequent steps.
2. Animal / Toxicology Studies: May be reduced or waived depending on the strength of the analytical similarity data. FDA judges on a case-by-case basis.
3. PK/PD Bridging Studies: Clinical pharmacology studies demonstrating equivalent absorption, distribution, and elimination.
4. Clinical Immunogenicity: Anti-drug antibody (ADA) studies to confirm the biosimilar does not induce new immune reactions compared to the reference.
5. Comparative Clinical Studies: May be waived if PK/PD and analytical data are sufficiently robust. Risk-based approach.
6. Interchangeability requires a showing that the biosimilar is expected to produce the same clinical result as the reference product in any given patient, with additional statutory considerations for products administered more than once. FDA determines what evidence is needed in each case; switching studies are not an automatic universal requirement.

10.4 Biologic Exclusivity and the Patent Dance

The BPCIA establishes a unique exclusivity clock for biologics that differs significantly from the Hatch-Waxman regime for small molecules:

Exclusivity Type	Duration	Description
4-Year Filing Bar	4 years	No biosimilar 351(k) application can even be filed within 4 years of the reference product's approval date.
12-Year Market Exclusivity	12 years	No biosimilar can receive FDA approval for 12 years from the reference product approval. The longest exclusivity period in pharmaceutical law.
Paediatric Extension	+6 months	Additional 6 months if the reference sponsor completes required paediatric studies.
Interchangeable Exclusivity	1 year	The first biosimilar to achieve interchangeable designation receives 1 year of exclusivity before another interchangeable biosimilar can be approved.

The BPCIA created a statutory framework for patent information exchanges and litigation involving biosimilars, commonly called the 'patent dance.' The precise duties and litigation consequences are governed by the statute and relevant case law; the exchange should not be described as an unconditional, universally mandatory negotiation process.

The Purple Book (purple.fda.gov) is the biologics equivalent of the Orange Book. It lists all licensed biological products, their biosimilar and interchangeable designations, and the reference product status of each.

> Example: Humira (adalimumab) is the reference product for all adalimumab biosimilars. Following the expiry of Humira's exclusivity period in 2023, multiple biosimilars launched in

the US market. Hadlima, Hyrimoz, Cyltezo, and others appear in the Purple Book as 351(k) biosimilars. Cyltezo received the first interchangeable designation for adalimumab.

Chapter 11: Quality Management Systems (QMS)

A Quality Management System is the operational backbone of pharmaceutical and medical device compliance. It is the totality of documented processes, procedures, responsibilities, and resources that an organisation uses to ensure that its products consistently meet regulatory requirements and quality standards. A robust QMS is not merely a regulatory obligation — it is the primary defence against FDA inspection findings, warning letters, and market withdrawal.

11.1 The Six Core QMS Pillars

11.1.1 Document Control

Every regulated document Standard Operating Procedures (SOPs), work instructions, forms, specifications must be controlled: approved before use, version-controlled, and accessible to those who need them while protected from unauthorised changes. Superseded versions must be archived and retrievable.

FDA inspection signal: If an investigator asks to see a specific SOP and the staff member retrieves an outdated version, or if the current version cannot be immediately located, this is a document control deficiency that will appear in a Form 483 observation.

11.1.2 Change Control

Any change to a product, process, material, equipment, or facility must be evaluated, approved, and implemented through a formal change control process. The change control record must document the nature of the change, the risk assessment, any necessary testing or validation, and the regulatory impact (e.g., does this change require a CBE-30 supplement or a PAS?).

11.1.3 CAPA — Corrective and Preventive Action

CAPA is the engine of continuous quality improvement. When a deviation, non-conformance, customer complaint, or audit finding occurs, a CAPA investigation systematically identifies the root cause, implements corrective actions (fixing the specific problem), and implements preventive actions (eliminating the root cause to prevent recurrence). Every CAPA must be closed with documented evidence that the actions were effective.

Example: A CAPA is opened after a batch of tablets fails dissolution testing. Root cause investigation reveals that a raw material supplier changed their manufacturing process without notification (a supplier qualification failure). Corrective action: reject the batch and re-test retained samples. Preventive action: update supplier qualification requirements to include change notification obligations, audit the supplier, and implement incoming material testing for this parameter.

11.1.4 Training Management

All personnel who perform activities affecting product quality must be trained in the relevant procedures before they perform those activities. Training records must document what was

trained, when, who delivered the training, and whether the trainee demonstrated competency. Re-training must be triggered whenever a procedure is updated.

11.1.5 Deviation Management

Deviations should be documented, assessed and investigated according to their risk, impact and applicable quality-system procedures. Root-cause analysis, CAPA and effectiveness checks are applied when warranted rather than automatically required in the same form for every minor departure.

11.1.6 Audit and Inspection Readiness

Internal audits conducted by personnel independent of the area being audited are the organisation's primary mechanism for verifying that its QMS is functioning as designed before an FDA inspector arrives. Supplier qualification audits extend the QMS requirements to critical vendors. Inspection readiness means maintaining continuous compliance, not a reactive scramble in the weeks before a scheduled inspection.

Chapter 12: Post-Market Surveillance — Report, Record, Recall

The regulatory journey does not end at product approval or device clearance. Post-market surveillance is the systematic process of collecting, evaluating, and acting on safety information that emerges during the real-world use of approved drugs and cleared or approved devices. This is where regulatory professionals spend the majority of their careers.

12.1 Report — Adverse Event Capture

Adverse-event reporting requirements depend on product category, reporter type, seriousness and reportability criteria. Manufacturers, importers and device user facilities have different mandatory device-reporting obligations. Patients and healthcare professionals may also submit voluntary reports. Suspected events can be reportable without establishing definitive causality, but not every observed event requires an FDA submission.

Product Type	Reporting Mechanism	Database	Timeline for Serious Unexpected AEs
NDA drugs, ANDA generics, BLA biologics	MedWatch Form 3500A (industry mandatory)	FAERS — FDA Adverse Event Reporting System	15 calendar days (expedited)
Medical devices (510k, De Novo, PMA)	Medical Device Report (MDR)	MAUDE — Manufacturer and User Facility Device Experience	30 days (5 days for urgent events)

Patients and healthcare providers can also voluntarily submit reports using MedWatch Form 3500 (without the 'A' suffix).

12.2 Record — Triage and Complaint File Management

Not every adverse event requires escalation to the FDA. A structured triage process determines the appropriate response:

- Routine or expected side effects (e.g., mild nausea with a chemotherapy agent): documented in the complaint file as an internal record. No escalation required unless frequency signals a new safety signal.
- Potentially reportable deaths, serious injuries and certain device malfunctions must be evaluated under the applicable MDR requirements. For example, device manufacturers generally report within 30 calendar days, with five-workday reporting in defined circumstances. User facilities generally report qualifying deaths or serious injuries within 10 workdays to the required recipients.
- Birth defects: automatic escalation pathway regardless of causality assessment.

The Minimal Viable Report for a medical device MDR requires four core elements:

- (1) Patient identifier (anonymised where required by privacy law)
- (2) The event itself: nature of the side effect, injury, or malfunction
- (3) Date of the event
- (4) Product identification: lot number, Universal Device Identifier (UDI), serial number, and trade name

Complaint and MDR records must satisfy applicable FDA recordkeeping requirements. For electronic records and signatures within Part 11's scope, appropriate controls must be implemented; not all records automatically trigger every Part 11 requirement.

12.3 Recall — Market Withdrawal

When a product presents a risk to the public due to safety signals, contamination, labeling errors, or manufacturing failures a recall process is initiated. Recalls are classified by severity:

Class	Risk Level	FDA Action
Class I	Highest urgency: immediate danger to public health. Probability of serious health consequences is high.	Urgent public notification. May include press release, healthcare provider letters, and immediate market withdrawal.
Class II	Adverse health consequences are possible but not immediate. Less severe or reversible harm.	Standard recall notification. Healthcare provider and patient communication.
Class III	Product violates FDA regulations but is unlikely to cause adverse health consequences.	Market correction: consumer-level notification typically not required.

> Example: In 2023, a manufacturer recalled a powered infusion pump (Class II recall) after receiving MDR reports that the pump's alarm system failed to alert nurses when the drug reservoir was depleted. While no deaths occurred, the potential for under-dosing critical medications was identified as a moderate health risk. The recall required hospitals to return affected units and apply a firmware update.

Chapter 13: FDA Adverse Event Monitoring Systems and MAUDE

Both FAERS and MAUDE are publicly searchable databases. This makes them not only regulatory reporting repositories, but also powerful competitive intelligence and risk assessment tools. Experienced regulatory professionals mine these databases proactively before a submission to understand known failure modes, anticipate FDA concerns, and benchmark their own safety profiles.

13.1 AEMS / Historical FAERS — Drug and Biologic Adverse Event Monitoring

FDA announced AEMS in March 2026 as part of its adverse-event data modernization initiative. Because interfaces, scope and data migration may evolve, consult FDA's current AEMS, FAERS and program-specific documentation before describing which system accepts or displays a particular product's reports.

FAERS can be searched to:

- Identify all reported adverse events for a competitor’s drug before designing your own safety monitoring plan
- Assess disproportionality signals (whether a specific AE is reported more frequently with your drug than would be expected by chance a pharmacovigilance technique called EBGM or PRR analysis)
- Review regulatory actions taken in response to safety signals such as labeling updates, REMS additions, or market withdrawals to understand FDA’s safety decision-making framework

> Example: A company preparing an NDA for a new JAK inhibitor searches FAERS for all adverse events reported with approved JAK inhibitors (tofacitinib, baricitinib, upadacitinib). They identify a consistent signal of opportunistic infections and major adverse cardiovascular events (MACE). This intelligence shapes the safety monitoring plan, the clinical stopping rules, and the proposed labeling for their drug before the first Phase 3 patient is enrolled.

13.2 MAUDE — Manufacturer and User Facility Device Experience

MAUDE is FDA’s searchable repository of medical device reports. Reporting is incomplete and information may be unverified. MAUDE can support signal detection and hazard identification, but the number of reports cannot, by itself, establish device failure rates, event incidence, causality or comparative safety.

MAUDE is particularly valuable for:

- 510(k) preparation: identifying known failure modes of the predicate device and demonstrating that your device mitigates or matches those risks
- Design risk analysis: using real-world device failure data to inform FMEA (Failure Mode and Effects Analysis)
- Post-market surveillance planning: identifying possible device failure modes and signals; obtaining valid denominator and exposure data separately before estimating rates.

Example: Before submitting a 510(k) for a new smart infusion pump, the regulatory team searches MAUDE for all MDRs filed against the predicate pump (product code FRN). They find 47 reports of alarm failure and 12 reports of motor stall. They design their bench testing protocol to specifically address alarm reliability and motor performance, and they describe these test results in the 510(k) to proactively demonstrate that their device does not share these failure modes.

Chapter 14: Language Hygiene — Writing Like a Regulatory Professional

Every word written in a regulatory document an MDR, a FAERS report, a 510(k) submission, a recall notice, a labeling claim carries legal and regulatory weight. Objective language protects the organisation. Subjective, conclusory, or emotionally loaded language invites regulatory scrutiny, legal liability, and often prolongs FDA review.

The Golden Rule: Write what you observed, not what you concluded. FDA reviewers, plaintiff attorneys, and foreign regulators will draw their own conclusions from your documented objective facts. Your job is to record, not to interpret causality or assign blame.

14.1 Drug Adverse Event (MedWatch / FAERS)

✕ Don't Write This	✓ Write This Instead
Our drug caused a headache in the patient.	Patient reported onset of headache (MedDRA term: headache) on [date], approximately 2 hours following oral administration of [drug name, lot number XXXXX]. Severity: mild. Action taken: no dose modification required. Outcome: resolved without intervention.

Why it matters: The phrase 'our drug caused' asserts legal causality a conclusion that belongs to the FDA's pharmacovigilance scientists, not the reporter. The objective report lets FDA's population-level signal detection determine causation from thousands of similar reports. Writing 'caused' in a report also creates litigation risk.

14.2 Device Malfunction (MDR / MAUDE)

✕ Don't Write This	✓ Write This Instead
The device failed and put the patient at serious risk of harm.	Device screen went blank during intravenous infusion of [drug name] at [facility name], [date]. Audible alarm did not activate. Infusion was suspended by the nurse upon observing the screen. No patient harm reported. Device serial number [XXXXX] returned to manufacturer for analysis.

Why it matters: 'The device failed' and 'put the patient at serious risk' are both conclusions, not observations. 'Screen went blank' is a factual observation. Describe precisely what occurred, in

what sequence, at what location, with what outcome. FDA can determine the malfunction and risk level from those facts.

14.3 510(k) Predicate Comparison

✕ Don't Write This	✓ Write This Instead
Our device is basically the same as the predicate.	The proposed device has the same intended use as predicate device K[XXXXXX]: controlled intravenous drug delivery in hospitalised adult patients. The proposed device and the predicate share substantially equivalent technological characteristics with respect to fluid pathway materials, pump mechanism (peristaltic), flow rate range (0.1–999.9 mL/hr), and alarm systems. The proposed device incorporates a colour touchscreen interface in place of the predicate's monochrome LCD. This difference does not raise new questions of safety or effectiveness because both provide equivalent information to the clinician; bench testing confirms equivalent alarm performance under IEC 60601-2-24.

Why it matters: 'Basically the same' has no regulatory meaning. Substantial Equivalence is a defined legal standard with specific criteria. Your predicate comparison language must mirror the legal definition precisely to satisfy FDA's reviewers.

14.4 Recall / Field Safety Notice

✕ Don't Write This	✓ Write This Instead
We are recalling this product due to a manufacturing defect that could cause serious harm.	Voluntary Recall Notice Class II. Affected product: [Product Name], Lot Numbers [XXXX–XXXX], Expiry [Date]. A non-conformance was identified during routine stability testing at month 12: dissolution rate for the active ingredient was observed at 68% at 45 minutes, below the specification minimum of 75%. Root cause: polymorphic shift of the active ingredient attributable to storage temperature excursion during distribution. Potential risk: sub-therapeutic drug exposure. Estimated probability of adverse health consequences: low. Action required: quarantine and return affected lots. Replacement product available within 14 business days. Contact [name, contact details] to arrange return.

Why it matters: Vague language in a recall notice creates confusion in the supply chain, delays the return of affected product, and fails to enable prescribers and patients to make informed decisions. Every sentence must specify the exact product, specific lots, precise non-conformance, risk level, and clear action.

Chapter 15: Regulatory Strategy — Connecting Commercial Goals to Evidence

Regulatory strategy is not merely compliance planning. It is the discipline of aligning a company's commercial objectives, its evidence generation plan, and the applicable regulatory requirements into a coherent, risk-managed development programme. A strategic regulatory professional thinks not only about what the FDA requires, but about how regulatory decisions drive the commercial value of the product.

15.1 The Unified Pathway Matrix

Every sound regulatory strategy rest on three pillars working in alignment:

Commercial Goal

What indication or therapeutic claim are you pursuing? What is the target patient population adults, paediatric patients, a rare disease population? The commercial goal defines the indication, which in turn defines what clinical evidence is required. A narrow indication may be faster to develop but limits commercial opportunity. A broad indication may have greater commercial value but requires a larger, more expensive clinical programme.

Development Risk Bridge

Map each regulatory requirement to the corresponding milestone in your development timeline. Identify the FDA meeting touchpoints that de-risk each transition: Pre-IND (confirm the development plan before Phase 1), End-of-Phase 2 (confirm Phase 3 design and statistical approach before committing to Phase 3), Pre-NDA / Pre-BLA (confirm dossier completeness before submission). Each FDA interaction reduces uncertainty and therefore reduces development risk.

Regulatory Requirement

Identify which FDA centre handles your product type and which specific review division is responsible. The regulatory pathway, evidence standards, and review expectations vary significantly across CDER (drugs), CBER (biologics, vaccines, gene therapy), and CDRH (medical devices). The centre assignment determines everything downstream.

> Critical insight: Never design a clinical trial before confirming the regulatory pathway. It is significantly cheaper to pay for a well-structured Pre-IND meeting than to run a clinical trial that generates data the FDA cannot use to support your application.

15.2 Expedited FDA Programmes for Unmet Medical Needs

These programmes do not replace the standard development milestone strip. They sit alongside it, providing additional FDA engagement, rolling review options, or priority review benefits for products that address serious conditions with unmet medical needs:

Programme	Eligibility	Key Benefit

Fast Track	Serious condition + unmet medical need	Enables rolling review (submit modules as completed). More frequent FDA meetings. Does not guarantee Priority Review.
Breakthrough Therapy	Preliminary clinical evidence of substantial improvement over existing therapy	Most intensive FDA guidance and interaction. Organising cross-disciplinary FDA review teams. Rolling review.
RMAT (Regenerative Medicine Advanced Therapy)	Cell therapy, tissue engineering, gene therapy for serious conditions	Combines Breakthrough benefits with tailored manufacturing flexibility for novel manufacturing approaches.
Orphan Designation	Rare disease affecting fewer than 200,000 patients in the US	7-year market exclusivity + 50% tax credit on clinical trial costs + filing fee waiver. Does not affect review pathway.

15.3 Centre-Specific Regulatory Playbooks

CDER (Center for Drug Evaluation and Research) — Drugs

The essential dos and don’ts for CDER submissions:

- Do: Check the Orange Book early to verify TE codes, patent status, and exclusivity blocks before investing in ANDA development
- Do: Account for formulation differences a different salt or ester form of the active ingredient typically requires new bioavailability clinical data
- Do: Use DRUGS@FDA to benchmark prior NDA review letters and Complete Response Letters before writing your own dossier
- Do: Match the correct 21 CFR Part for your submission type (Part 314 for NDA, Part 320 for bioequivalence, Parts 600 series for biologics)
- Do Not: Assume that a generic drug requires zero clinical data sometimes bioequivalence bridging studies are needed even for well-established molecules
- Do Not: Ignore exclusivity blocks the 3-year exclusivity, 5-year NCE exclusivity, 7-year orphan exclusivity, and 6-month paediatric extension can all delay generic launch
- Do Not: File an NDA without a Pre-NDA meeting the risk of an unexpected major deficiency in a complete response letter is substantial

CBER (Center for Biologics Evaluation and Research) — Biologics

- Do: Take the 12-year market exclusivity clock seriously biosimilar launch strategy and financial modelling both depend on it
- Do: Apply the 4-letter non-proprietary name suffix rule (e.g., adalimumab-atto) FDA requires biosimilars to carry a distinguishing suffix
- Do: Front-load the analytical similarity package for 351(k) submissions strong structural and functional characterisation can reduce or eliminate the need for comparative clinical studies
- Do: Navigate the Purple Book to confirm reference product status and check interchangeable designations before defining your biosimilar development strategy

- Do Not: Treat biologics like small molecule generics. ‘Highly similar’ is not equivalent to ‘identical.’ The biosimilar characterisation package is orders of magnitude more complex than a generic’s BE study
- Do Not: Assume interchangeable status automatically follows biosimilar approval. Interchangeability requires a separate, higher-burden demonstration
- Do Not: Treat immunogenicity assessment as unimportant. The evidence required for a biosimilar is determined using a risk-based, totality-of-the-evidence approach; the need for a comparative clinical immunogenicity study depends on the program and current FDA guidance.

CDRH (Center for Devices and Radiological Health) — Medical Devices

- Do: Find the magic key search the FDA Product Classification Database for the exact 3-letter product code and regulation number before writing a single page of a 510(k)
- Do: Conduct thorough predicate homework. Understand the distinction between intended use (what the device does clinically) and technological characteristics (how it works technically) before selecting your predicate
- Do: Request a Pre-Submission (Q-Sub) meeting with FDA before filing major submissions. This is free, and FDA’s written feedback significantly reduces the risk of additional information requests
- Do: Search MAUDE for known failure modes of your predicate device before designing your bench testing protocol
- Do Not: Over-file or rely on a weak predicate. A poorly supported substantial equivalence argument creates a bad submission. If the predicate doesn’t hold, consider De Novo
- Do Not: Use the term ‘approved’ for a 510(k)-cleared device. FDA ‘clears’ 510(k) devices and ‘approves’ PMA devices. These terms are legally distinct, and their misuse is noted by FDA
- Do Not: Use an impermissible split-predicate argument by relying on one device for intended use and an unrelated device for technological characteristics. FDA may allow multiple predicates in defined circumstances when substantial equivalence is supported.

Key Regulatory References and Resources

The following FDA resources are referenced throughout this document and should be bookmarked by all regulatory affairs professionals:

- Orange Book: orangebook.fda.gov
- DRUGS@FDA: accessdata.fda.gov/scripts/cder/daf
- Purple Book: purple.fda.gov
- FDA AEMS / adverse-event monitoring: https://www.fda.gov/drugs/fda-adverse-event-monitoring-system-aems
- MAUDE Database: accessdata.fda.gov/scripts/cdrh/cfdocs/cfmaude
- FDA Product Classification Database (devices): accessdata.fda.gov/scripts/cdrh/cfdocs/cfpcd
- Electronic Submissions Gateway (ESG): esgproxy.fda.gov
- FDA 21 CFR Online Database: ecfr.gov/current/title-21

- ICH M4 Guideline (eCTD structure): ich.org
- BPCIA (Public Health Service Act Section 351): relevant USC sections on FDA website